5. System Response and LTI Properties

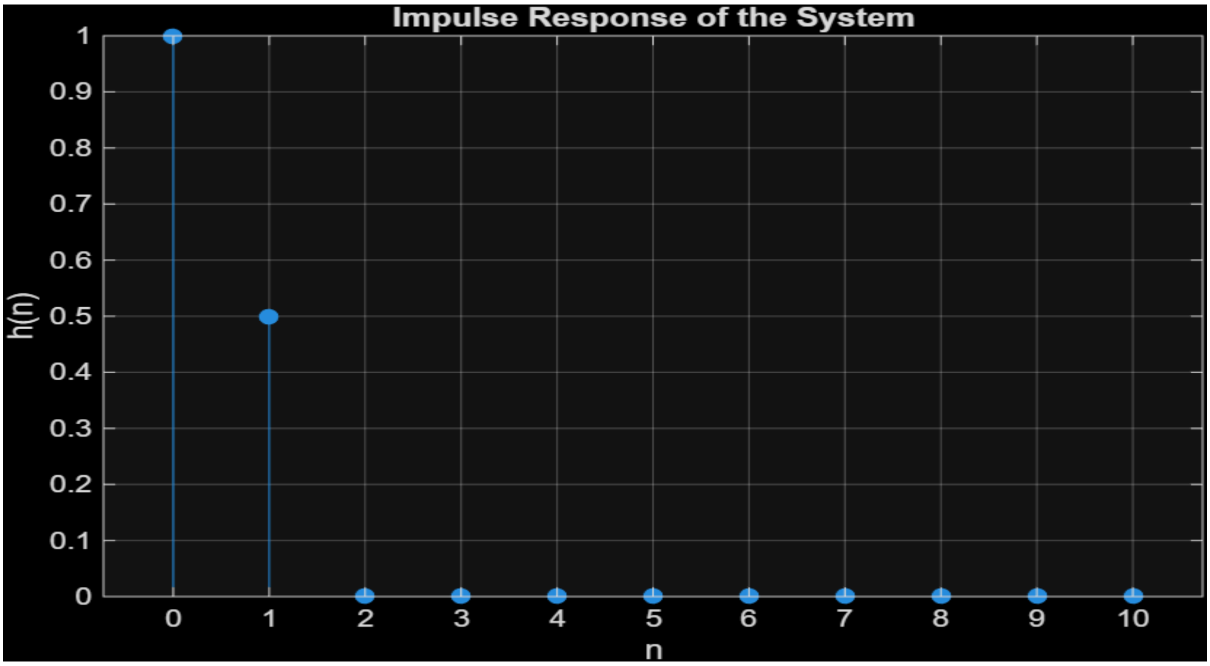
Experiment 5.1:

Impulse Response of the System

Program:

```
clc;
clear;
close all;
% Sample index
n = 0:10;
% Unit impulse input
x = [1 zeros(1,10)];
% System: y(n)=x(n)+0.5x(n-1)
b = [1 0.5];
a = [1];
% Impulse response
h = filter(b,a,x);
% Display result
disp('Impulse Response h(n):');
disp(h);
% Plot
figure;
stem(n,h,'filled');
grid on;
title('Impulse Response of the System');
xlabel('n');
ylabel('h(n)');
```

Output Waveform:



Result:

The impulse response of the discrete-time system y(n)=x(n)+0.5x(n−1) was obtained by applying a unit impulse signal as input. The resulting impulse response was h(n)={1,0.5,0,0,…}, which characterizes the behavior of the system.

Experiment 5.2:

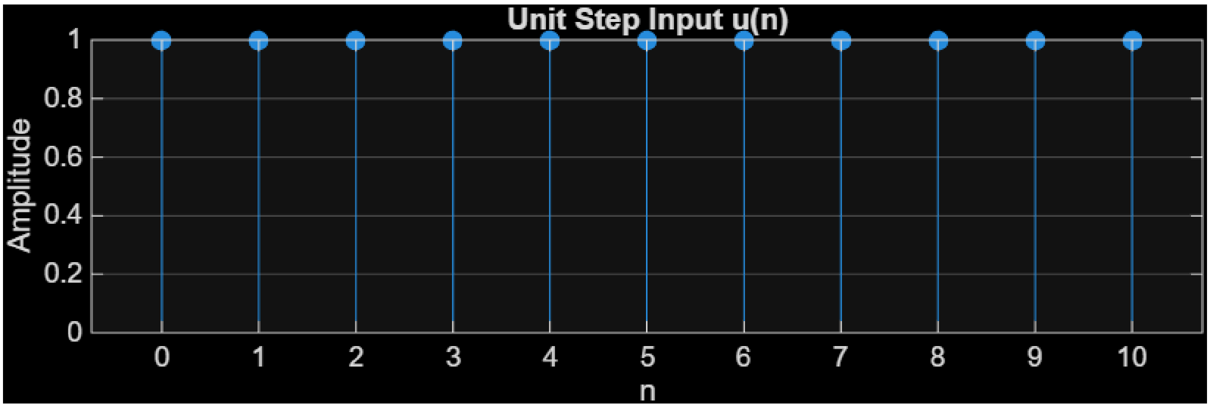
Step Response of the System

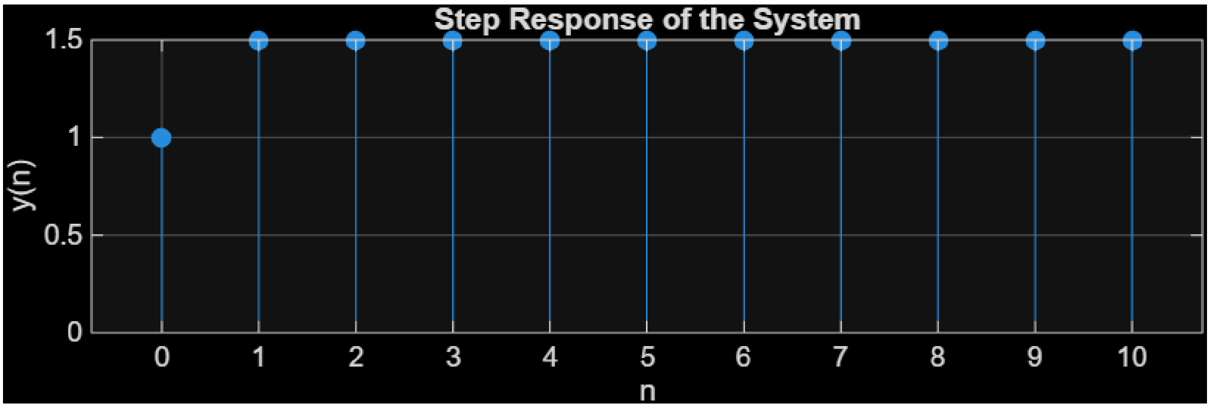
Program:

```
clc;
clear;
close all;
% Sample index
n = 0:10;
% Unit step input
x = ones(1,length(n));
% System: y(n)=x(n)+0.5x(n-1)
b = [1 0.5];
a = [1];
% Step response
y = filter(b,a,x);
% Display result
```

```
disp('Step Response y(n):');
disp(y);
% Plot
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Unit Step Input u(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Step Response of the System');
xlabel('n');
ylabel('y(n)');
```

Output Waveform:





Result:

The step response of the discrete-time system $y(n)=x(n)+0.5x(n-1)$ was obtained by applying a unit step input. The output reached a steady-state value of 1.5, demonstrating the response of the system to a constant input signal.